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SEAT WITH REMOVABLE HEADREST

Abstract

A seat of a vehicle can include a removable headrest with a locking and release mechanism and a trimmed bun region which engages with armatures of the seat that extend from a seat back. The user can remove the headrest by pressing a release button on a side of the headrest. The user can then lift up to slide the headrest from the armatures. The headrest can then be stored within the vehicle interior (e.g., glovebox, back on the seat, on the seat cushion) or frunk for safe storage. One or more sensors (e.g., interior camera) can be operated to ensure that the headrest is removed from the seat prior to folding operation. One or more sensors (e.g., interior camera) can be operated to ensure that the headrest is installed onto the seat prior to one or more operations of the vehicle.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This application claims the benefit of priority to U.S. Provisional Patent Application No. 63/560,593, entitled, "SEAT WITH REMOVABLE HEADREST", filed on Mar. 1, 2024, the disclosure of which is hereby incorporated herein in its entirety.

INTRODUCTION
[0002] The present description relates generally to vehicles. More particularly, the present disclosure relates to management systems for a seat with a removable headrest. Vehicles are often provided with seats that can be adjusted with various positions and orientations to accommodate the preferences of a variety of users.
SUMMARY
[0003] Embodiments of the present disclosure are directed toward a seat of a vehicle that can include a removable headrest with a locking and release mechanism and a trimmed bun region which engages with armatures of the seat that extend from a seat back. The user can remove the headrest by pressing a release button on a side of the headrest. The user can then lift up to slide the headrest from the armatures. The headrest can then be stored within the vehicle interior (e.g., glovebox, back on the seat, on the seat cushion) or frunk for safe storage. One or more sensors (e.g., interior camera) can be operated to ensure that the headrest is removed from the seat prior to folding operation. One or more sensors (e.g., interior camera) can be operated to ensure that the headrest is installed onto the seat prior to one or more operations of the vehicle.
[0004] In accordance with aspects of the subject technology, an apparatus is provided that includes a seat back; an armature extending from a portion of the seat back and comprising a striker; and a headrest configured to receive a portion of the armature. The headrest includes a latch configured to releasably engage the striker and a release button configured to actuate the latch to release the striker.
[0005] The latch may be configured to rotate within the headrest to transition between a locked configuration and an unlocked configuration. The latch may be biased to the unlocked configuration. The headrest may further include a lever configured to maintain the latch in the locked configuration while the lever is in a first configuration, wherein the release button is configured to actuate the latch by transitioning the lever to a second configuration, in which the lever is configured to allow the latch to transition to the unlocked configuration. The release button may define a portion of an outer periphery of the headrest. The portion of the armature may be a first portion of the armature, wherein the seat back is configured to receive a second portion of the armature. A portion of the latch may extend from the headrest to within a portion of the seat back to releasably engage the striker.
[0006] In accordance with aspects of the subject technology, a method is provided that includes: receiving a transition signal comprising an instruction to transition a seat from an extended configuration to a folded configuration; and in response to receiving the transition signal: detecting a configuration of a headrest of the seat; in accordance with a determination that the headrest is attached to the seat, outputting a notification; and in accordance with a determination that the headrest is detached from the seat, transitioning the seat from the extended configuration to the folded configuration.
[0007] The method may further include, after outputting the notification: detecting the

configuration of the headrest of the seat; in accordance with the determination that the headrest is attached to the seat, maintaining output of the notification; and in accordance with the determination that the headrest is detached from the seat, ceasing output of the notification and transitioning the seat from the extended configuration to the folded configuration.

[0008] The method may further include, in response to receiving the transition signal and in accordance with the determination that the headrest is attached to the seat, forgoing transition of the seat from the extended configuration to the folded configuration. Detecting the configuration of the headrest of the seat may include operating a camera to optically detect whether the headrest is attached to the seat. The notification may include an instruction to a user to detach the headrest from the seat. The transition signal may be generated in response to a user input at an input device of the seat.

[0009] In accordance with aspects of the subject technology, a method is provided that includes: receiving a start signal comprising an instruction to perform an operation of a vehicle; and in response to receiving the start signal: detecting a configuration of a headrest of a seat of the vehicle; in accordance with a determination that the headrest is attached to the seat, performing the operation of the vehicle; and in accordance with a determination that the headrest is detached from the seat, forgoing performance of the operation of the vehicle.

[0010] The method may further include, in response to receiving the start signal and in accordance with the determination that the headrest is detached from the seat, outputting a notification. The method may further include: after outputting the notification and while forgoing performance of the operation of the vehicle, receiving an override signal from a user; and in response to receiving the override signal from the user, performing the operation. The override signal may be generated in response to a user input at an input device of the vehicle. The notification may include an instruction to a user to attach the headrest to the seat. The operation may include operating a motor of the vehicle. Detecting the configuration of the headrest of the seat may include operating a camera to optically detect whether the headrest is attached to the seat.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0011] Certain features of the subject technology are set forth in the appended claims. However, for purpose of explanation, several embodiments of the subject technology are set forth in the following figures.

[0012] FIG. **1** illustrates a schematic perspective side view of an example implementation of a vehicle having a dashboard in accordance with one or more implementations.

[0013] FIG. **2** illustrates a perspective view of an example implementation of a seat in accordance with one or more implementations.

[0014] FIG. **3** illustrates a perspective view of an example implementation of a portion of a seat in accordance with one or more implementations.

[0015] FIG. **4** illustrates a side view of an example implementation of a seat with a removable headrest in accordance with one or more implementations.

[0016] FIG. **5** illustrates a perspective view of an example implementation of a headrest in accordance with one or more implementations.

[0017] FIG. **6** illustrates a perspective view of an example implementation of a dashboard for a vehicle having containers in a closed configuration in accordance with one or more implementations.

[0018] FIG. **7** illustrates a perspective view of an example implementation of the dashboard of FIG. **2** having containers in an open configuration in accordance with one or more implementations.

[0019] FIG. **8** illustrates a side view of an example implementation of a seat in a folded

configuration and a dashboard in accordance with one or more implementations.
[0020] FIG. **9** illustrates a side view of an example implementation of a seat in a folded configuration and a dashboard in accordance with one or more implementations.
[0021] FIG. **10** illustrates a schematic view of an example implementation of a locking assembly for a headrest of a seat with a latch in a locked configuration in accordance with one or more implementations.
[0022] FIG. **11** illustrates a schematic view of an example implementation of the locking assembly of FIG. **10** seat with a button in an actuated configuration in accordance with one or more implementations.
[0023] FIG. **12** illustrates a schematic view of an example implementation of the locking assembly of FIGS. **10** and **11** with the latch in an unlocked configuration in accordance with one or more implementations.
[0024] FIG. **13** illustrates a schematic view of an example implementation of a locking assembly for a headrest of a seat with a latch in a locked configuration in accordance with one or more implementations.
[0025] FIG. **14** illustrates a schematic view of an example implementation of the locking assembly of FIG. **13** seat with a button in an actuated configuration and the latch in an unlocked configuration in accordance with one or more implementations.
[0026] FIG. **15** illustrates a schematic view of an example implementation of the locking assembly of FIGS. **13** and **14** with the headrest removed from the seat in accordance with one or more implementations.
[0027] FIG. **16** illustrates a schematic view of an example implementation of a locking assembly for a headrest of a seat with a latch in a locked configuration in accordance with one or more implementations.
[0028] FIG. **17** illustrates a schematic view of an example implementation of the locking assembly of FIG. **16** seat with a button in an actuated configuration and the latch in an unlocked configuration in accordance with one or more implementations.
[0029] FIG. **18** illustrates a schematic view of an example implementation of the locking assembly of FIGS. **16** and **17** with the headrest removed from the seat in accordance with one or more implementations.
[0030] FIG. **19** illustrates a flow diagram of an example process for managing a locking system based on one or more detections in accordance with one or more implementations of the subject technology.
[0031] FIG. **20** illustrates a flow diagram of an example process for managing a locking system based on one or more detections in accordance with one or more implementations of the subject technology.
[0032] FIG. **21** illustrates an electronic system with which one or more implementations of the subject technology may be implemented.

DETAILED DESCRIPTION

[0033] The detailed description set forth below is intended as a description of various configurations of the subject technology and is not intended to represent the only configurations in which the subject technology can be practiced. The appended drawings are incorporated herein and constitute a part of the detailed description. The detailed description includes specific details for the purpose of providing a thorough understanding of the subject technology. However, the subject technology is not limited to the specific details set forth herein and can be practiced using one or more other implementations. In one or more implementations, structures and components are shown in block diagram form in order to avoid obscuring the concepts of the subject technology.
[0034] Embodiments of the present disclosure are directed toward a management system for one or more seats with a removable headrest. Apparatuses, such as vehicles, buildings, and/or other enclosed and/or indoor spaces are often provided with seats. Such seats can support one or more

users.
[0035] Some seats can provide a capability to at least partially fold. However, with some seats (e.g., front seats), the dashboard and other features (e.g., steering wheel) of the vehicle provide insufficient clearance between the and the headrests of the folding seats during the folding motion. Where headrests are removable, the seat can be provided with additional clearance. However, the removable portion of some headrests are excessively large, which the requires a larger space for storage thereof until it is returned to the seat.
[0036] Embodiments of the present disclosure include a headrest of a seat with a locking and release mechanism and a trimmed bun region which engages with armatures of the seat that extend from a seat back. The user can remove the headrest by pressing a release button on a side of the headrest. The user can then lift up to slide the headrest from the armatures. The headrest can then be stored within the vehicle interior (e.g., glovebox, back on the seat, on the seat cushion) or frunk for safe storage. One or more sensors (e.g., interior camera) can be operated to ensure that the headrest is removed from the seat prior to folding operation. This ensures that the seat does not clash with the dashboard, steering wheel, or other features of the vehicle. One or more sensors (e.g., interior camera) can be operated to ensure that the headrest is installed onto the seat prior to one or more operations of the vehicle.
[0037] FIG. **1** is a diagram illustrating an example implementation of an apparatus as described herein. In the example of FIG. **1**, the apparatus is a moveable apparatus implemented as a vehicle **100**. In one or more implementations, the vehicle **100** may be implemented as an electric vehicle and may include one or more batteries for powering the vehicle and/or one or more systems and/or components of the vehicle.
[0038] For example, in one or more implementations, the vehicle **100** may be an electric vehicle having one or more electric motors that drive the wheels **102** of the vehicle using electric power from the battery. In one or more implementations, the vehicle **100** may also, or alternatively, include one or more chemically powered engines, such as a gas-powered engine or a fuel cell powered motor. For example, electric vehicles can be fully electric or partially electric (e.g., hybrid or plug-in hybrid).
[0039] In the example of FIG. **1**, the vehicle **100** is implemented as a truck (e.g., a pickup truck) having a dashboard **20** within a passenger compartment **110** of the vehicle **100**. The vehicle **100** can further include one or more seats **50** within the passenger compartment **110**. While only one seat **50** is illustrated in FIG. one, it will be understood that any number of seats **50** can be provided within the passenger compartment **110** of the vehicle **100**. For example, the seats **50** can be provided in one or more rows within the passenger compartment **110**. At least some of the seats **50** can have one or more features in common while being positioned at different locations within the passenger compartment **110**. One or more of the seats **50** can be positioned near a corresponding one of the doors **106** of the vehicle **100**. Accordingly, the doors **106** can provide access to the passenger compartment **110** and one or more of the seats. Anyone or more of the seats **50** can be provided with the features described herein.
[0040] The one or more seats **50** can be movable with respect to the dashboard **20** and/or one or more other structures within the passenger compartment **110**. As shown, the vehicle **100** may also include processing circuitry **108** (e.g., one or more processors, memory, and/or communications circuitry). The processing circuitry **108** may be communicatively coupled to the seats **50** and/or one or more other components of the vehicle **100**, for control, by the processing circuitry **108**, thereof.
[0041] As examples, the processing circuitry **108** of the vehicle **100** may include one or more processors (e.g., single processors, multi-core processors, central processing units (CPUs), application-specific integrated circuits (ASICS), field programmable gate arrays (FPGAs) and/or other processing circuits), and/or any of various types of computer-readable and/or machine-readable media (e.g., persistent storage, system memory and/or buffers, volatile memory and/or non-volatile memory). In one or more implementations, the processing circuitry **108** may include

input devices, output devices, network interfaces, and/or a bus that communicatively couples the processor(s), the memory, the communications circuitry, the input devices, the output devices, and/or one or more other devices or components (e.g., seats **50**, displays, cameras, motion sensors, proximity sensors, etc.). The processor(s) of the processing circuitry **108** may execute instructions stored in the memory of the processing circuitry **108**, such as to execute hardware, firmware, and/or software processes in order to perform the processes of the subject disclosure.

[0042] The example of FIG. **1**, in which the vehicle **100** is implemented as a pickup truck having a truck bed, is merely illustrative. In other implementations, the vehicle **100** may implemented as another type of electric truck, an electric delivery van, an electric automobile, an electric car, an electric motorcycle, an electric scooter, an electric passenger vehicle, an electric passenger or commercial truck, a hybrid vehicle, or other vehicles such as sea or air transport vehicles, planes, helicopters, submarines, boats, or drones, and/or any other movable apparatus having a dashboard **20**, one or more seats **50**, and processing circuitry **108**.

[0043] In one or more implementations, the seats **50** and/or processing circuitry **108** as described herein may also, or alternatively, be implemented in another apparatus, such as a building (e.g., a residential home or commercial building, or any other building) or other stationary apparatus.

[0044] Referring now to FIGS. **2-5**, one or more seats of the vehicle can provide an ability to releasably engage and remove a headrest. It will be understood that the features described herein with regard to an illustrative example of a seat can be applied to any one or more seats of a vehicle.

[0045] As shown in FIG. **2**, a seat **50** can include a seat base **52**, a seat back **54**, and a headrest **60**. The seat base **52** can form a portion upon which a user may be seated while operating or traveling in the vehicle. The seat base **52** can extend generally horizontally for supporting the user against the force of gravity. The seat back **54** can extend generally vertically in at least one configuration thereof for supporting the user against a force from forward acceleration of the vehicle. The headrest **60** can be positioned at an upper region of the seat back **54**. The headrest **60** can support the user against a force from forward acceleration of the vehicle.

[0046] As shown in FIG. **3**, the seat **50** can include one or more mechanisms for coupling the headrest **60** to the seat back. For example, the seat back can include one or more armatures **56** extending from an upper portion thereof. In some embodiments, the seat back can form a shield **55** that extends vertically alongside the armatures **56**. The shield **55** can extend at least as far as the armatures **56**, such that the shield **55** defined a farthest extent of the seat back. As further shown in FIG. **3**, the headrest **60** can include a release button **62**. The release button **62** can be provided on a lateral side of the headrest **60**. For example, the release button **62** can on a side of the headrest **60** that faces a corresponding door of the vehicle.

[0047] As shown in FIG. **4**, a headrest **60** can include a recessed portion **64** that is configured to face and receive a portion of a shield of a seat back. The recessed portion **64** can be on a rear side of the headrest **60**. The headrest **60** can further define one or more openings **66** for receiving the armatures of the seat back. The headrest **60** can include one or more mechanisms within the opening **66** for engaging the armatures, as described further herein.

[0048] As shown in FIG. **5**, the headrest **60** can be removed from the seat back **54**. In some embodiments, the release button **62** of the headrest **60** can be actuated to facilitate and/or otherwise allow removal of the headrest **60**. While or after the release button **62** is actuated, the headrest **60** can be removed from the armatures **56**.

[0049] In some embodiments, one or more portions of the seat **50** can be moved with respect to the other portions and/or other features of the vehicle. In some embodiments, the seat **50** can be adjusted translationally and/or rotationally. For example, the seat base **52** can be configured to move within one or more axes and/or rotate about one or more axes. By further example, the seat back **54** can be configured to move within one or more axes and/or rotate about one or more axes. The seat back **54** can form an angle **57** with respect to the seat base **52** and/or a floor of the vehicle. The seat back **54** can be adjustable within a range of angles **57**, such as 0-180 degrees. The seat **50** can include or be coupled to one or more actuators **51** that facilitate such adjustments. An actuator **51** may include one or more devices for directly or indirectly moving and/or controlling the seat **50** and/or components thereof. The actuator **51** may be one or more types of actuators such as an electric, magnetic, mechanical, or any other type of actuator. The actuator **51** can include motors, hydraulic actuators, pneumatic actuators, magnetic actuators, piezoelectric actuators, electroactive materials, stepper motors, shape-memory alloys, and the like, as well as drivetrain components such as gears, clutches, and/or transmissions, to facilitate movement of the seat **50** and/or its components based on operation of the actuator **51**.

[0050] In some embodiments, the seat **50** can include an input device **40** for receiving user inputs. Such user inputs can include generate signals that include instructions to adjust one or more features of the seat **50**. The input device **40** can be provided at the seat **50** and/or any location of the vehicle. An input device **40** can be operated to initiate transition or one, more than one, or all of the seats. Accordingly, a single input can be provide cause transition of multiple seats **50**.

[0051] Referring now to FIGS. **6** and **7**, a container system **10** can be located within a passenger compartment of a vehicle. As shown in FIG. **6**, the vehicle can include a dashboard **20** that includes and/or supports one or more components, such as a steering wheel **22** and/or a display device **24** configured to detect a user input. In some embodiments, the display device **24** and/or one or more other user interface devices can provide an ability to receive inputs from a user. In some embodiments, the display device **24** and/or one or more other user interface devices can provide an ability to provide outputs to a user.

[0052] As shown in FIG. **6**, the dashboard **20** and/or another portion of the vehicle can include a sensor **26**. The sensor **26** can be operated to detect one or more conditions within the passenger compartment. For example, the sensor **26** can detect a condition of a seat, a headrest, a user, and/or one or more other conditions with respect to the vehicle. In some embodiments, the sensor **26** can include a camera, optical sensor, depth sensor, proximity sensor, and the like. In some embodiments, the sensor **26** can be integrated within a component with respect to which the sensor **26** is operated to detect a condition. For example, the sensor **26** can be integrated within a seat for detections related to the seat, as described further herein.

[0053] In some embodiments, one or more containers **30** and **34** can be located along a dashboard **20** of the vehicle. For example, one or more containers **30** and **34** can be positioned at one or more sides of and/or beneath a steering wheel **22** and/or a display device **24** at the dashboard **20**. In some embodiments, the one or more containers **30** and **34** can be aligned along a common side (e.g., underside) of the dashboard **20**. In some embodiments, separate containers **30** and **34** can be positioned adjacent to each other.

[0054] As shown in FIG. **7**, the dashboard **20** can define one or more recesses for receiving the containers **30** and **34**. Each of the containers **30** and **34** can transition between a closed configuration (FIG. **6**) (e.g., within the recesses of the dashboard **20**) and an open configuration (FIG. **7**) (e.g., extending away from the dashboard **20**). The seats of the vehicle (not shown) can facilitate transition of the containers between the open configuration and the closed configuration, as described further herein.

[0055] Referring now to FIGS. **8** and **9**, a headrest of one or more seats can be removed to facilitate transitions and/or adjustments of the one or more seats. As shown in FIG. **8**, the headrest (not shown) can be removed from the seat back **54** of a first seat **50**A. In the depicted example, the first seat **50**A can be a driver seat that is aligned with the steering wheel **22** extending from the dashboard **20**. Removal of the headrest can leave the shield **55** and/or the armatures **56** of the seat back **54** defining an uppermost extent of the seat back **54**. As such, the overall height of the seat back **54** is reduced following removal of the headrest. Accordingly, the seat back **54** is provided with greater clearance with respect to the steering wheel **22** and/or other components of the dashboard **20**. In such a configuration, the transition of the seat back **54** to a folded configuration is enhanced by providing clearance with respect to other structures of the vehicle. It will be understood that such adjustments from an extended configuration to the folded the configuration can include other adjustments to the first seat **50**A, such as a change in the position and/or orientation of the seat base **52** (e.g., away from the steering wheel **22** and/or other components of the dashboard **20**).

[0056] As shown in FIG. **9**, the headrest **60** can remain attached to the seat back **54** of a second seat **50**B. Where such a second seat **50**B is at another location, the dashboard **20** can provide adequate clearance even when the headrest **60** remains attached to the seat back **54**. As such, the second seat **50**B can undergo transitions to the folded configuration without requiring removal of the headrest **60**. It will be understood that such adjustments from an extended configuration to the folded the configuration can include other adjustments to the seat **50**B, such as a change in the position and/or orientation of the seat base **52** (e.g., away from the dashboard **20**).

[0057] Referring now to FIGS. **10-12**, a headrest can provide one or more locking mechanisms for releasably securing to armatures of a seat back. As shown in FIG. **10**, a seat **50** can include a headrest **60** that is releasably attachable to one or more armatures **56** extending from a seat back (not shown). The one or more armatures **56** can include a striker **58**, for example at a terminal end thereof. The striker **58** can be received within an opening **66** of the headrest **60**. In a locked configuration, a latch **80** of the headrest **60** can engage the striker **58**, for example to prevent removal thereof from the opening **66** until release of the latch **80**. For example, at least a portion of the latch **80** can be interposed between the striker **58** and the opening **66**, such that removal of the latch **80** from such a location is required before the striker **58** can be removed from the opening **66**. In some embodiments, the latch **80** can be maintained in the locked configuration by one or more mechanisms, such as a lever **70**. The release button **62** of the headrest **60** can be operable (e.g., by a user) to actuate the lever **70** and/or the latch **80**. In some embodiments, the latch **80** can be biased to move away from the striker **58** but can be retained in a locked configuration by the lever **70**.

[0058] As shown in FIG. **11**, the release button **62** can be actuated, for example by a force applied by a user. Actuation of the release button **62** can result in actuation of the lever **70**. In some embodiments, the lever **70** can rotate about a lever pivot **72**. Additionally or alternatively, the lever **70** can move translationally. By actuating the lever **70**, the lever **70** can be moved with respect to (e.g., away from) the latch **80**. In some embodiments, the seat **50** can include a sensor **26**. The sensor can be located at the seat back (e.g., at the one or more armatures **56**) and/or at the headrest **60**. The sensor **26** can be operated to detect a condition of the seat **50**, the headrest **60**, a user, and/or one or more other conditions with respect to the vehicle. In some embodiments, the sensor **26** can include a camera, optical sensor, depth sensor, proximity sensor, and the like. In some embodiments, the sensor **26** can be operated to detect whether the headrest **60** is attached to the seat back (e.g., at the one or more armatures **56**). For example, the sensor **26** can be located at the one or more armatures **56** and detect the presence and/or engagement of one or more components of the headrest **60**. By further example, the sensor **26** can be located at the headrest **60** and detect the presence and/or engagement of one or more components of the seat back (e.g., one or more armatures **56**).

[0059] As shown in FIG. **12**, as the lever **70** is moved with respect to (e.g., away from) the latch **80**, the latch **80** can be permitted to move with respect to the striker **58**. In some embodiments, the latch **80** can rotate about a latch pivot **82**. Additionally or alternatively, the latch **80** can move translationally. In some embodiments, the latch **80** can be biased to move away from the striker **58**, and removal of the lever **70** can allow the latch **80** to move to the unlocked configuration thereof. With the latch **80** in the unlocked configuration, the striker **58** can be permitted to pass out of the opening **66**, which can further allow the headrest **60** to be released and removed from the armatures **56**. Where the latch **80** is biased to the unlocked configuration, the latch **80** can be maintained in the unlocked configuration even when the user input (e.g., force) is removed from the release button **62**.

[0060] In some embodiments, as shown in FIGS. **10-12**, the seat **50** includes a linkage of parts in an assembly that includes the release button **62**, the lever **70**, and/or the latch **80**, such that physical actuation of the release button **62** (e.g., by a user) causes and/or otherwise results in physical actuation of the lever **70** and/or the latch **80**. It should be understood that one or more other types of inputs can cause and/or otherwise result in physical actuation of the lever **70** and/or the latch **80**, for example via one or more intermediate mechanisms. In some embodiments, the release button **62** can be an actively operated input device, such as an electromechanical switch, an electrical button, a touchscreen, and the like. A user input at the release button **62** can activate one or more mechanisms for controlling and/or effecting physical actuation of the lever **70** and/or the latch **80**. In some embodiments, the seat **50** includes one or more motors that can be operated to actuate the lever **70** and/or the latch **80** in response to a user input at the release button **62**. In some embodiments, the seat **50** includes one or more shape memory alloys (e.g., at the lever **70** and/or the latch **80**) and a thermal output device (e.g., heater, etc.) that is operable to control a temperature of the one or more shape memory alloys and correspondingly actuate the lever **70** and/or the latch **80** in response to a user input at the release button **62**. In some embodiments, the seat **50** includes one or more magnets (e.g., at the lever **70** and/or the latch **80**) and one or more solenoids and/or other magnetic field generators that are operable to influence the one or more magnets and correspondingly actuate the lever **70** and/or the latch **80** in response to a user input at the release button **62**.

[0061] The headrest can be secured to the armatures **56** by inserting the striker **58** into the opening until it reaches the latch **80**. Further advancement can cause the striker **58** to actuate the latch **80** to the locked configuration (FIG. **10**). In the absence of a user input (e.g., force) at the release button **62**, the latch **80** can be maintained in the locked configuration (e.g., by the lever **70**).

[0062] Referring now to FIGS. **13-15**, a seat back can provide one or more locking mechanisms for releasably securing to armatures of a headrest. As shown in FIG. **13**, a seat **150** can include a headrest **160** that includes and/or is connected or attachable to one or more armatures **156** extending from a body thereof. The one or more armatures **156** can include a striker **170** (e.g., a notch, indentation, ledge, shoulder, recess, detent, protrusion, hook, pin, and/or combinations thereof). The armatures **156** can be received within respective housings **190** of the seat back **154**, for example within openings or channels thereof. The housings **190** can be accessible from an exterior of the seat back **154** while being at and/or recessed within an outer periphery of a body of the seat back **154**. The armatures **156** can be inserted until the strikers **170** are aligned with respective latches **180** of the seat back **154**. In a locked configuration, the latches **180** of the seat back **154** can engage the respective strikers **170**, for example to prevent removal thereof from the housings **190** until release of the latch **180**. For example, at least a portion of each latch **180** can be interposed between the respective striker **170** and a respective entry of the housings **190**, such that removal of the latch **180** from such a location (e.g., at the strikers **170**) is required before the armatures **156** (e.g., including the strikers **170**) can be removed from the housings **190**. In some embodiments, the latch **180** can be maintained in the locked configuration by one or more mechanisms, such as a spring bias (not shown). The release button **162** of the seat back **154** can be operable (e.g., by a user) to actuate the latches **180**.

[0063] In some embodiments, the seat **150** can include a sensor (not shown), such as the sensor **26** of FIGS. **10-12**. The sensor can be located at the seat back **154** (e.g., at the one or more armatures **156**) and/or at the headrest **160**. The sensor can be operated to detect a condition of the seat **150**, the headrest **160**, a user, and/or one or more other conditions with respect to the vehicle. In some embodiments, the sensor can include a camera, optical sensor, depth sensor, proximity sensor, and the like. In some embodiments, the sensor can be operated to detect whether the headrest **160** is attached to the seat back **154** (e.g., at the one or more armatures **156**).

[0064] As shown in FIG. **14**, the release button **162** can be actuated, for example by a force applied by a user. Actuation of the release button **162** can result in actuation of the latches **180**. In some embodiments, the latches **180** can translate along with the release button **162**. Additionally or alternatively, the latches **180** can move rotationally and/or translationally in a manner that is different from the movement of the release button **162**. By actuating the latches **180**, the latches **180** can be moved with respect to (e.g., away from) the strikers **170**.

[0065] In some embodiments, as shown in FIG. **14**, the seat **150** includes a linkage of parts in an assembly that includes the latches **180** and the release button **162**, such that physical actuation of the release button **162** (e.g., by a user) causes and/or otherwise results in physical actuation of the one or more latches **180**. It should be understood that one or more other types of inputs can cause and/or otherwise result in physical actuation of the one or more latches **180**, for example via one or more intermediate mechanisms. In some embodiments, the release button **162** can be an actively operated input device, such as an electromechanical switch, an electrical button, a touchscreen, and the like. A user input at the release button **162** can activate one or more mechanisms for controlling and/or effecting physical actuation of the one or more latches **180**. In some embodiments, the seat **150** includes one or more motors that can be operated to actuate the one or more latches **180** in response to a user input at the release button **162**. In some embodiments, the seat **150** includes one or more shape memory alloys (e.g., at the one or more latches **180**) and a thermal output device (e.g., heater, etc.) that is operable to control a temperature of the one or more shape memory alloys and correspondingly actuate the one or more latches **180** in response to a user input at the release button **162**. In some embodiments, the seat **150** includes one or more magnets (e.g., at the one or more latches **180**) and one or more solenoids and/or other magnetic field generators that are operable to influence the one or more magnets and correspondingly actuate the one or more latches **180** in response to a user input at the release button **162**.

[0066] As shown in FIG. **15**, as the latches **180** are moved with respect to (e.g., away from) the strikers **170**, the headrest **160** (e.g., including the armatures **156**) can be permitted to move with respect to the seat back **154**. For example, with the latches **180** in the unlocked configuration, the armatures **156** can be permitted to pass out of the housings **190**, which can further allow the headrest **160** to be released and removed from the seat back **154**.

[0067] The headrest **160** can be stored (e.g., along with the armatures **156**) in another location, such as a glove box. The headrest **160** can be secured again to the seat back **154** by inserting the armatures **156** into the respective housings **190**. Inserting the armatures **156** can result in the strikers **170** coming into alignment and engagement with the latches **180**. In the absence of a user input (e.g., force) at the release button **162**, the latches **180** can be maintained in the locked configuration to retain the headrest **160** with respect to the seat back **154**.

[0068] Referring now to FIGS. **16-18**, a headrest can provide one or more locking mechanisms for releasably securing to a headrest. As shown in FIG. **16**, a seat **250** can include a headrest **260** that includes and/or is connected or attachable to one or more armatures **256** extending from a body thereof. The armatures **256** can be received within respective housings **290** of the seat back **254**, for example within openings or channels thereof. The housings **290** can be accessible from an exterior of the seat back **254** while being at and/or recessed within an outer periphery of a body of the seat back **254**. The housings **290** can each include a striker **292** (e.g., a notch, indentation, ledge, shoulder, recess, detent, protrusion, hook, pin, and/or combinations thereof). The armatures **256** can be inserted until the latches **280** of the headrest **260** are aligned with respective strikers **292** of the seat back **254**. In a locked configuration, the latches **280** of the headrest **260** can engage the respective strikers **292**, for example to prevent removal of the armatures **256** from the housings **290** until release of the latches **280**. For example, at least a portion of each striker **292** can be interposed between the respective latch **280** and a respective entry of the housings **290**, such that removal of the latch **280** from such a location (e.g., at the strikers **292**) is required before the armatures **256** can be removed from the housings **290**. In some embodiments, the latch **280** can be maintained in the locked configuration by one or more mechanisms, such as a spring bias (not shown). The release button **262** of the headrest **260** can be operable (e.g., by a user) to actuate the latches **280**. In some embodiments, the release button of the headrest **260** can be connected to the latches **280**, which can extend to the seat back **254** (e.g., at least to the strikers **292**). Accordingly, engagement can occur within the seat back **254** while release control can be provided at the headrest **260**.

[0069] In some embodiments, the seat **250** can include a sensor (not shown), such as the sensor **26** of FIGS. **10-12**. The sensor can be located at the seat back **254** (e.g., at the one or more armatures **256**) and/or at the headrest **260**. The sensor can be operated to detect a condition of the seat **250**, the headrest **260**, a user, and/or one or more other conditions with respect to the vehicle. In some embodiments, the sensor can include a camera, optical sensor, depth sensor, proximity sensor, and the like. In some embodiments, the sensor can be operated to detect whether the headrest **260** is attached to the seat back **254** (e.g., at the one or more armatures **256**).

[0070] As shown in FIG. **17**, the release button **262** can be actuated, for example by a force applied by a user. Actuation of the release button **262** can result in actuation of the latches **280**. In some embodiments, the latches **280** can translate along with the release button **262**. Additionally or alternatively, the latches **280** can move rotationally and/or translationally in a manner that is different from the movement of the release button **262**. For example, as shown in FIG. **16**, the release button **262** can be moved in a first direction (e.g., away from the seat back **254**), and the latches **280** can be moved in one or more other directions, different from the first direction. By further example, by actuating the latches **280**, the latches **280** can be moved with respect to (e.g., away from) the strikers **292** (e.g., towards each other as shown in FIG. **17**).

[0071] In some embodiments, as shown in FIG. **17**, the seat **250** includes a linkage of parts in an assembly that includes the latches **280** and the release button **262**, such that physical actuation of the release button **262** (e.g., by a user) causes and/or otherwise results in physical actuation of the one or more latches **280**. It should be understood that one or more other types of inputs can cause and/or otherwise result in physical actuation of the one or more latches **280**, for example via one or more intermediate mechanisms. In some embodiments, the release button **262** can be an actively operated input device, such as an electromechanical switch, an electrical button, a touchscreen, and the like. A user input at the release button **262** can activate one or more mechanisms for controlling and/or effecting physical actuation of the one or more latches **280**. In some embodiments, the seat **250** includes one or more motors that can be operated to actuate the one or more latches **280** in response to a user input at the release button **262**. In some embodiments, the seat **250** includes one or more shape memory alloys (e.g., at the one or more latches **280**) and a thermal output device (e.g., heater, etc.) that is operable to control a temperature of the one or more shape memory alloys and correspondingly actuate the one or more latches **280** in response to a user input at the release button **262**. In some embodiments, the seat **250** includes one or more magnets (e.g., at the one or more latches **280**) and one or more solenoids and/or other magnetic field generators that are operable to influence the one or more magnets and correspondingly actuate the one or more latches **280** in response to a user input at the release button **262**.

[0072] As shown in FIG. **18**, as the latches **280** are moved with respect to (e.g., away from) the strikers **292**, the headrest **260** (e.g., including the armatures **256**) can be permitted to move with respect to the seat back **254**. For example, with the latches **280** in the unlocked configuration, the armatures **256** can be permitted to pass out of the housings **290**, which can further allow the headrest **260** to be released and removed from the seat back **254**.

[0073] The headrest **260** can be stored (e.g., along with the armatures **256**) in another location, such as a glove box. The headrest **260** can be secured again to the seat back **254** by inserting the armatures **256** into the respective housings **290**. Inserting the armatures **256** can result in the strikers **292** coming into alignment and engagement with the latches **280**. In the absence of a user input (e.g., force) at the release button **262**, the latches **280** can be maintained in the locked configuration to retain the headrest **260** with respect to the seat back **254**.

[0074] FIG. **19** illustrates a flow diagram of an example process **1300** for managing transition of a seat in response to a user input in accordance with one or more implementations of the subject technology. For explanatory purposes, the process **1300** is primarily described herein with reference to the vehicle **100**, seat **50**, headrest **60**, seat **150**, headrest **160**, seat **250**, and/or headrest **260** of FIGS. **1-18**, and/or various components thereof. However, the process **1300** is not limited to the vehicle **100**, seat **50**, headrest **60**, seat **150**, headrest **160**, seat **250**, and/or headrest **260** of FIGS. **1-18**, and one or more blocks (or operations) of the process **1300** may be performed by one or more other structural components of the vehicle **100** and/or of other suitable moveable apparatuses, devices, or systems. Further, for explanatory purposes, some of the blocks of the process **1300** are described herein as occurring in serial, or linearly. However, multiple blocks of the process **1300** may occur in parallel. In addition, the blocks of the process **1300** need not be performed in the order shown and/or one or more blocks of the process **1300** need not be performed and/or can be replaced by other operations.

[0075] At block **1302**, the vehicle can receive a transition signal comprising an instruction to transition a seat. For example, the transition signal can include an instruction to transition a seat from an extended configuration to a folded configuration. The transition signal can be provided by a user via an input device, such as a control on a portion of the seat, a display device, and/or another user interface device (e.g., at an interior region and/or an exterior region of the vehicle). In some embodiments, the transition signal includes an instruction to transition multiple seats (e.g., all seats of the vehicle and/or a row or other portion of the vehicle) to a folded configuration. Accordingly, the configuration of multiple seats can be modified and/or otherwise controlled in response to a single input. In some embodiments, the transition signal is generated in response to user input (e.g., at an input device). In some embodiments, the transition signal is generated in response to a user input at a seat. In some embodiments, the transition signal is generated in response to a user input at another portion of the vehicle, such as a liftgate. In some embodiments, the transition signal is generated in response to a detected configuration (e.g., open or closed) of another portion of the vehicle, such as a liftgate.

[0076] At block **1304**, the vehicle can detect a headrest configuration. For example, the vehicle can detect whether a headrest is attached to a seat back of the seat or detached from the seat back. In some embodiments, the vehicle can detect the configuration of the headrest by operating a sensor, such as a camera providing a field of view that includes a seat. In some embodiments, the vehicle can detect the configuration of the headrest by operating a sensor at an interface between the headrest and the seat back. In some embodiments, headrest detection can be limited to those seats for which removal of the headrest is required to complete the transition (e.g., the seat of FIG. **8**, rather than the seat of FIG. **9**). In some embodiments, where headrest removal is not required to complete the transition, headrest detection can be omitted and transitions can be performed based on whether or not headrests are present for seats for which removal is required.

[0077] At block **1306**, if the vehicle detects that the headrest is detached from (e.g., is not attached to) the seat back, the process **1300** can proceed to block **1308**. If the vehicle detects that the headrest is not detached from (e.g., is attached to) the seat back, the process **1300** can proceed to block **1310**.

[0078] At block **1308**, the vehicle can transition the seat from the extended configuration to folded configuration. For example, upon detecting that the headrest is detached from the seat back, the vehicle can determine that adequate clearance is provided to transition the seat. In some embodiments, the vehicle can adjust the seat in one or more other ways. For example, the vehicle can adjust the position and/or orientation of one or more portions of the seat. It will be understood that such operations can be provided in series or in parallel.

[0079] Following block **1308**, the process **1300** can return to block **1302** and/or another block and/or process.

[0080] At block **1310**, the vehicle can forgo transition of the seat and/or can output a notification. For example, the vehicle can output a notification via a display device, a speaker, and/or one or more other user interface devices of the vehicle. In some embodiments, the notification can include an instruction to remove the headrest. Following block **1310**, the process **1300** can return to block

1304 and/or another block and/or process.

[0081] FIG. **20** illustrates a flow diagram of an example process **1400** for managing an operation in response to a user input in accordance with one or more implementations of the subject technology. For explanatory purposes, the process **1400** is primarily described herein with reference to the vehicle **100**, seat **50**, headrest **60**, seat **150**, headrest **160**, seat **250**, and/or headrest **260** of FIGS. **1-18**, and/or various components thereof. However, the process **1400** is not limited to the vehicle **100**, seat **50**, headrest **60**, seat **150**, headrest **160**, seat **250**, and/or headrest **260** of FIGS. **1-18**, and one or more blocks (or operations) of the process **1400** may be performed by one or more other structural components of the vehicle **100** and/or of other suitable moveable apparatuses, devices, or systems. Further, for explanatory purposes, some of the blocks of the process **1400** are described herein as occurring in serial, or linearly. However, multiple blocks of the process **1400** may occur in parallel. In addition, the blocks of the process **1400** need not be performed in the order shown and/or one or more blocks of the process **1400** need not be performed and/or can be replaced by other operations.

[0082] At block **1402**, the vehicle can receive a start signal comprising an instruction to start an operation. For example, the start signal can include an instruction to transition a seat from an extended configuration to a folded configuration. The start signal can be provided by a user via an input device, such as a start/stop button, a steering wheel, a pedal, a display device, and/or another user interface device.

[0083] At block **1404**, the vehicle can detect a headrest configuration. For example, the vehicle can detect whether a headrest is attached to a seat back of the seat or detached from the seat back. In some embodiments, the vehicle can detect the configuration of the headrest by operating a sensor, such as a camera providing a field of view that includes a seat. In some embodiments, the vehicle can detect the configuration of the headrest by operating a sensor at an interface between the headrest and the seat back. In some embodiments, headrest detection can be limited to those seats at which a user is present (e.g., headrest detection is preceded by user detection). In some embodiments, where no user is present, headrest detection can be omitted and operations can be performed based on whether or not headrests are present for seats containing users.

[0084] At block **1406**, if the vehicle detects that the headrest is attached to (e.g., is not detached from) the seat back, the process **1400** can proceed to block **1408**. If the vehicle detects that the headrest is not attached to (e.g., is detached from) the seat back, the process **1400** can proceed to block **1410**.

[0085] At block **1408**, the vehicle can perform one or more operations corresponding to the start signal. For example, upon detecting that the headrest is attached to the seat back, the vehicle can operate a motor of the vehicle and/or allow operation of a steering wheels and/or pedals of the vehicle.

[0086] Following block **1408**, the process **1400** can return to block **1402** and/or another block and/or process.

[0087] At block **1410**, the vehicle can forgo performance of the operation and/or output a notification. For example, the vehicle can output a notification via a display device, a speaker, and/or one or more other user interface devices of the vehicle. In some embodiments, the notification can include an instruction to install the headrest. Following block **1410**, the process **1400** can return to block **1404** and/or another block and/or process.

[0088] In some embodiments, the vehicle can receive an override signal comprising an instruction to perform the operation even if the headrest is not attached to (e.g., is detached from) the seat back. The override signal can be provided by a user via an input device, such as a start/stop button, a steering wheel, a pedal, a display device, and/or another user interface device. At block **1412**, if the vehicle does not receive the override signal, the process **1400** can proceed to block **1408**. If the vehicle receives the override signal, the process **1400** can return to block **1404** and/or another block and/or process.

[0089] FIG. **21** illustrates an example electronic system **1500** with which aspects of the present disclosure may be implemented. The electronic system **1500** can be, and/or can be a part of, any electronic device for providing the features and performing processes described in reference to FIGS. **1-14**, including but not limited to a vehicle, computer, server, smartphone, and wearable device (e.g., authentication device). The electronic system **1500** may include various types of computer-readable media and interfaces for various other types of computer-readable media. The electronic system **1500** includes a persistent storage device **1502**, system memory **1504** (and/or buffer), input device interface **1506**, output device interface **1508**, sensor(s) **1510**, ROM **1512**, processing unit(s) **1514**, network interface **1516**, bus **1518**, and/or subsets and variations thereof.
[0090] The bus **1518** collectively represents all system, peripheral, and chipset buses that communicatively connect the numerous internal devices and/or components of the electronic system **1500**, such as any of the components of the vehicle **100** discussed above with respect to FIG. **1**. In one or more implementations, the bus **1518** communicatively connects the one or more processing unit(s) **1514** with the ROM **1512**, the system memory **1504**, and the persistent storage device **1502**. From these various memory units, the one or more processing unit(s) **1514** retrieves instructions to execute and data to process in order to execute the processes of the subject disclosure. The one or more processing unit(s) **1514** can be a single processor or a multi-core processor in different implementations. In one or more implementations, one or more of the processing unit(s) **1514** may be included in a container system, such as in the form of the processing circuitry **108**.
[0091] The ROM **1512** stores static data and instructions that are needed by the one or more processing unit(s) **1514** and other modules of the electronic system **1500**. The persistent storage device **1502**, on the other hand, may be a read-and-write memory device. The persistent storage device **1502** may be a non-volatile memory unit that stores instructions and data even when the electronic system **1500** is off. In one or more implementations, a mass-storage device (such as a magnetic or optical disk and its corresponding disk drive) may be used as the persistent storage device **1502**.
[0092] In one or more implementations, a removable storage device (such as a floppy disk, flash drive, and its corresponding disk drive) may be used as the persistent storage device **1502**. Like the persistent storage device **1502**, the system memory **1504** may be a read-and-write memory device. However, unlike the persistent storage device **1502**, the system memory **1504** may be a volatile read-and-write memory, such as RAM. The system memory **1504** may store any of the instructions and data that one or more processing unit(s) **1514** may need at runtime. In one or more implementations, the processes of the subject disclosure are stored in the system memory **1504**, the persistent storage device **1502**, and/or the ROM **1512**. From these various memory units, the one or more processing unit(s) **1514** retrieves instructions to execute and data to process in order to execute the processes of one or more implementations.
[0093] The persistent storage device **1502** and/or the system memory **1504** may include one or more machine learning models. Machine learning models, such as those described herein, are often used to form predictions, solve problems, recognize objects in image data, and the like. For example, machine learning models described herein may be used to predict whether an authorized user is approaching a vehicle and intends to open a charging port closure. Various implementations of the machine learning model are possible. For example, the machine learning model may be a deep learning network, a transformer-based model (or other attention-based models), a multi-layer perceptron or other feed-forward networks, neural networks, and the like. In various examples, machine learning models may be more adaptable as machine learning models may be improved over time by re-training the models as additional data becomes available.
[0094] The bus **1518** also connects to the input device interfaces **1506** and output device interfaces **1508**. The input device interface **1506** enables a user to communicate information and select commands to the electronic system **1500**. Input devices that may be used with the input device

interface **1506** may include, for example, alphanumeric keyboards, touch screens, and pointing devices. The output device interface **1508** may enable the electronic system **1500** to communicate information to users. For example, the output device interface **1508** may provide the display of images generated by electronic system **1500**. Output devices that may be used with the output device interface **1508** may include, for example, printers and display devices, such as a liquid crystal display (LCD), a light emitting diode (LED) display, an organic light emitting diode (OLED) display, a flexible display, a flat panel display, a solid state display, a projector, or any other device for outputting information.
[0095] One or more implementations may include devices that function as both input and output devices, such as a touchscreen. In these implementations, feedback provided to the user can be any form of sensory feedback, such as visual feedback, auditory feedback, or tactile feedback; and input from the user can be received in any form, including acoustic, speech, or tactile input.
[0096] The bus **1518** also connects to sensor(s) **1510**. The sensor(s) **1510** may include a location sensor, which may be used in determining device position based on positioning technology. For example, the location sensor may provide for one or more of GNSS positioning, wireless access point positioning, cellular phone signal positioning, Bluetooth signal positioning, image recognition positioning, and/or an inertial navigation system (e.g., via motion sensors such as an accelerometer and/or gyroscope). In one or more implementations, the sensor(s) **1510** may be utilized to detect movement, travel, and orientation of the electronic system **1500**. For example, the sensor(s) may include an accelerometer, a rate gyroscope, and/or other motion-based sensor(s). The sensor(s) **1510** may include one or more biometric sensors and/or image sensors for authenticating a user.
[0097] The bus **1518** also couples the electronic system **1500** to one or more networks and/or to one or more network nodes through the one or more network interface(s) **1516**. In this manner, the electronic system **1500** can be a part of a network of computers (such as a local area network or a wide area network). Any or all components of the electronic system **1500** can be used in conjunction with the subject disclosure.
[0098] Implementations within the scope of the present disclosure can be partially or entirely realized using a tangible computer-readable storage medium (or multiple tangible computer-readable storage media of one or more types) encoding one or more instructions. The tangible computer-readable storage medium also can be non-transitory in nature.
[0099] The computer-readable storage medium can be any storage medium that can be read, written, or otherwise accessed by a general purpose or special purpose computing device, including any processing electronics and/or processing circuitry capable of executing instructions. For example, without limitation, the computer-readable medium can include any volatile semiconductor memory, such as RAM, DRAM, SRAM, T-RAM, Z-RAM, and TTRAM. The computer-readable medium also can include any non-volatile semiconductor memory, such as ROM, PROM, EPROM, EEPROM, NVRAM, flash, nvSRAM, FeRAM, FeTRAM, MRAM, PRAM, CBRAM, SONOS, RRAM, NRAM, racetrack memory, FJG, and Millipede memory.
[0100] Further, the computer-readable storage medium can include any non-semiconductor memory, such as optical disk storage, magnetic disk storage, magnetic tape, other magnetic storage devices, or any other medium capable of storing one or more instructions. In one or more implementations, the tangible computer-readable storage medium can be directly coupled to a computing device, while in other implementations, the tangible computer-readable storage medium can be indirectly coupled to a computing device, e.g., via one or more wired connections, one or more wireless connections, or any combination thereof.
[0101] Instructions can be directly executable or can be used to develop executable instructions. For example, instructions can be realized as executable or non-executable machine code or as instructions in a high-level language that can be compiled to produce executable or non-executable machine code. Further, instructions also can be realized as or can include data. Computer-

executable instructions also can be organized in any format, including routines, subroutines, programs, data structures, objects, modules, applications, applets, functions, etc. As recognized by those of skill in the art, details including, but not limited to, the number, structure, sequence, and organization of instructions can vary significantly without varying the underlying logic, function, processing, and output.

[0102] While the above discussion primarily refers to microprocessor or multi-core processors that execute software, one or more implementations are performed by one or more integrated circuits, such as ASICs or FPGAs. In one or more implementations, such integrated circuits execute instructions that are stored on the circuit itself.

[0103] A reference to an element in the singular is not intended to mean one and only one unless specifically so stated, but rather one or more. For example, “a” module may refer to one or more modules. An element proceeded by “a,” “an,” “the,” or “said” does not, without further constraints, preclude the existence of additional same elements.

[0104] Headings and subheadings, if any, are used for convenience only and do not limit the invention. The word exemplary is used to mean serving as an example or illustration. To the extent that the term include, have, or the like is used, such term is intended to be inclusive in a manner similar to the term comprise as comprise is interpreted when employed as a transitional word in a claim. Relational terms such as first and second and the like may be used to distinguish one entity or action from another without necessarily requiring or implying any actual such relationship or order between such entities or actions.

[0105] Phrases such as an aspect, the aspect, another aspect, some aspects, one or more aspects, an implementation, the implementation, another implementation, some implementations, one or more implementations, an embodiment, the embodiment, another embodiment, some embodiments, one or more embodiments, a configuration, the configuration, another configuration, some configurations, one or more configurations, the subject technology, the disclosure, the present disclosure, other variations thereof and alike are for convenience and do not imply that a disclosure relating to such phrase(s) is essential to the subject technology or that such disclosure applies to all configurations of the subject technology. A disclosure relating to such phrase(s) may apply to all configurations, or one or more configurations. A disclosure relating to such phrase(s) may provide one or more examples. A phrase such as an aspect or some aspects may refer to one or more aspects and vice versa, and this applies similarly to other foregoing phrases.

[0106] A phrase “at least one of” preceding a series of items, with the terms “and” or “or” to separate any of the items, modifies the list as a whole, rather than each member of the list. The phrase “at least one of” does not require selection of at least one item; rather, the phrase allows a meaning that includes at least one of any one of the items, and/or at least one of any combination of the items, and/or at least one of each of the items. By way of example, each of the phrases “at least one of A, B, and C” or “at least one of A, B, or C” refers to only A, only B, or only C; any combination of A, B, and C; and/or at least one of each of A, B, and C.

[0107] It is understood that the specific order or hierarchy of steps, operations, or processes disclosed is an illustration of exemplary approaches. Unless explicitly stated otherwise, it is understood that the specific order or hierarchy of steps, operations, or processes may be performed in different order. Some of the steps, operations, or processes may be performed simultaneously. The accompanying method claims, if any, present elements of the various steps, operations or processes in a sample order, and are not meant to be limited to the specific order or hierarchy presented. These may be performed in serial, linearly, in parallel or in different order. It should be understood that the described instructions, operations, and systems can generally be integrated together in a single software/hardware product or packaged into multiple software/hardware products.

[0108] In one aspect, a term coupled or the like may refer to being directly coupled. In another aspect, a term coupled or the like may refer to being indirectly coupled.

[0109] Terms such as top, bottom, front, rear, side, horizontal, vertical, and the like refer to an arbitrary frame of reference, rather than to the ordinary gravitational frame of reference. Thus, such a term may extend upwardly, downwardly, diagonally, or horizontally in a gravitational frame of reference.
[0110] The disclosure is provided to enable any person skilled in the art to practice the various aspects described herein. In some instances, well-known structures and components are shown in block diagram form in order to avoid obscuring the concepts of the subject technology. The disclosure provides various examples of the subject technology, and the subject technology is not limited to these examples. Various modifications to these aspects will be readily apparent to those skilled in the art, and the principles described herein may be applied to other aspects.
[0111] All structural and functional equivalents to the elements of the various aspects described throughout the disclosure that are known or later come to be known to those of ordinary skill in the art are expressly incorporated herein by reference and are intended to be encompassed by the claims. Moreover, nothing disclosed herein is intended to be dedicated to the public regardless of whether such disclosure is explicitly recited in the claims. No claim element is to be construed under the provisions of 35 U.S.C. § 112(f), unless the element is expressly recited using the phrase “means for” or, in the case of a method claim, the element is recited using the phrase “step for”.
[0112] Those of skill in the art would appreciate that the various illustrative blocks, modules, elements, components, methods, and algorithms described herein may be implemented as hardware, electronic hardware, computer software, or combinations thereof. To illustrate this interchangeability of hardware and software, various illustrative blocks, modules, elements, components, methods, and algorithms have been described above generally in terms of their functionality. Whether such functionality is implemented as hardware or software depends upon the particular application and design constraints imposed on the overall system. Skilled artisans may implement the described functionality in varying ways for each particular application. Various components and blocks may be arranged differently (e.g., arranged in a different order, or partitioned in a different way) all without departing from the scope of the subject technology.
[0113] The title, background, brief description of the drawings, abstract, and drawings are hereby incorporated into the disclosure and are provided as illustrative examples of the disclosure, not as restrictive descriptions. It is submitted with the understanding that they will not be used to limit the scope or meaning of the claims. In addition, in the detailed description, it can be seen that the description provides illustrative examples and the various features are grouped together in various implementations for the purpose of streamlining the disclosure. The method of disclosure is not to be interpreted as reflecting an intention that the claimed subject matter requires more features than are expressly recited in each claim. Rather, as the claims reflect, inventive subject matter lies in less than all features of a single disclosed configuration or operation. The claims are hereby incorporated into the detailed description, with each claim standing on its own as a separately claimed subject matter.
[0114] The claims are not intended to be limited to the aspects described herein, but are to be accorded the full scope consistent with the language of the claims and to encompass all legal equivalents. Notwithstanding, none of the claims are intended to embrace subject matter that fails to satisfy the requirements of the applicable patent law, nor should they be interpreted in such a way.

Claims

1. An apparatus comprising: a seat back; an armature extending from a portion of the seat back and comprising a striker; a headrest configured to receive a portion of the armature and comprising: a latch configured to releasably engage the striker; and a release button configured to actuate the latch to release the striker; and a sensor configured to detect whether the headrest is attached to the

seat back.
2. The apparatus of claim 1, wherein the latch is configured to rotate within the headrest to transition between a locked configuration and an unlocked configuration.
3. The apparatus of claim 2, wherein the latch is biased to the unlocked configuration.
4. The apparatus of claim 3, wherein the headrest further comprises a lever configured to maintain the latch in the locked configuration while the lever is in a first configuration, wherein the release button is configured to actuate the latch by transitioning the lever to a second configuration, in which the lever is configured to allow the latch to transition to the unlocked configuration.
5. The apparatus of claim 1, wherein the release button defines a portion of an outer periphery of the headrest.
6. The apparatus of claim 1, wherein the portion of the armature is a first portion of the armature, wherein the seat back is configured to receive a second portion of the armature.
7. The apparatus of claim 1, wherein a portion of the latch extends from the headrest to within a portion of the seat back to releasably engage the striker.
8. A method comprising: receiving a transition signal comprising an instruction to transition a seat from an extended configuration to a folded configuration; and in response to receiving the transition signal: detecting a configuration of a headrest of the seat; in accordance with a determination that the headrest is attached to the seat, outputting a notification; and in accordance with a determination that the headrest is detached from the seat, transitioning the seat from the extended configuration to the folded configuration.
9. The method of claim 8, further comprising, after outputting the notification: detecting the configuration of the headrest of the seat; in accordance with the determination that the headrest is attached to the seat, maintaining output of the notification; and in accordance with the determination that the headrest is detached from the seat, ceasing output of the notification and transitioning the seat from the extended configuration to the folded configuration.
10. The method of claim 8, further comprising, in response to receiving the transition signal and in accordance with the determination that the headrest is attached to the seat, forgoing transition of the seat from the extended configuration to the folded configuration.
11. The method of claim 8, wherein detecting the configuration of the headrest of the seat comprises operating a camera to optically detect whether the headrest is attached to the seat.
12. The method of claim 8, wherein the notification includes an instruction to a user to detach the headrest from the seat.
13. The method of claim 8, wherein the transition signal is generated in response to a user input at an input device of the seat.
14. A method comprising: receiving a start signal comprising an instruction to perform an operation of a vehicle; and in response to receiving the start signal: detecting a configuration of a headrest of a seat of the vehicle; in accordance with a determination that the headrest is attached to the seat, performing the operation of the vehicle; and in accordance with a determination that the headrest is detached from the seat, forgoing performance of the operation of the vehicle.
15. The method of claim 14, further comprising, in response to receiving the start signal and in accordance with the determination that the headrest is detached from the seat, outputting a notification.
16. The method of claim 15, further comprising: after outputting the notification and while forgoing performance of the operation of the vehicle, receiving an override signal from a user; and in response to receiving the override signal from the user, performing the operation.
17. The method of claim 16, wherein the override signal is generated in response to a user input at an input device of the vehicle.
18. The method of claim 15, wherein the notification includes an instruction to a user to attach the headrest to the seat.
19. The method of claim 14, wherein the operation comprises operating a motor of the vehicle.

20. The method of claim 14, wherein detecting the configuration of the headrest of the seat comprises operating a camera to optically detect whether the headrest is attached to the seat.